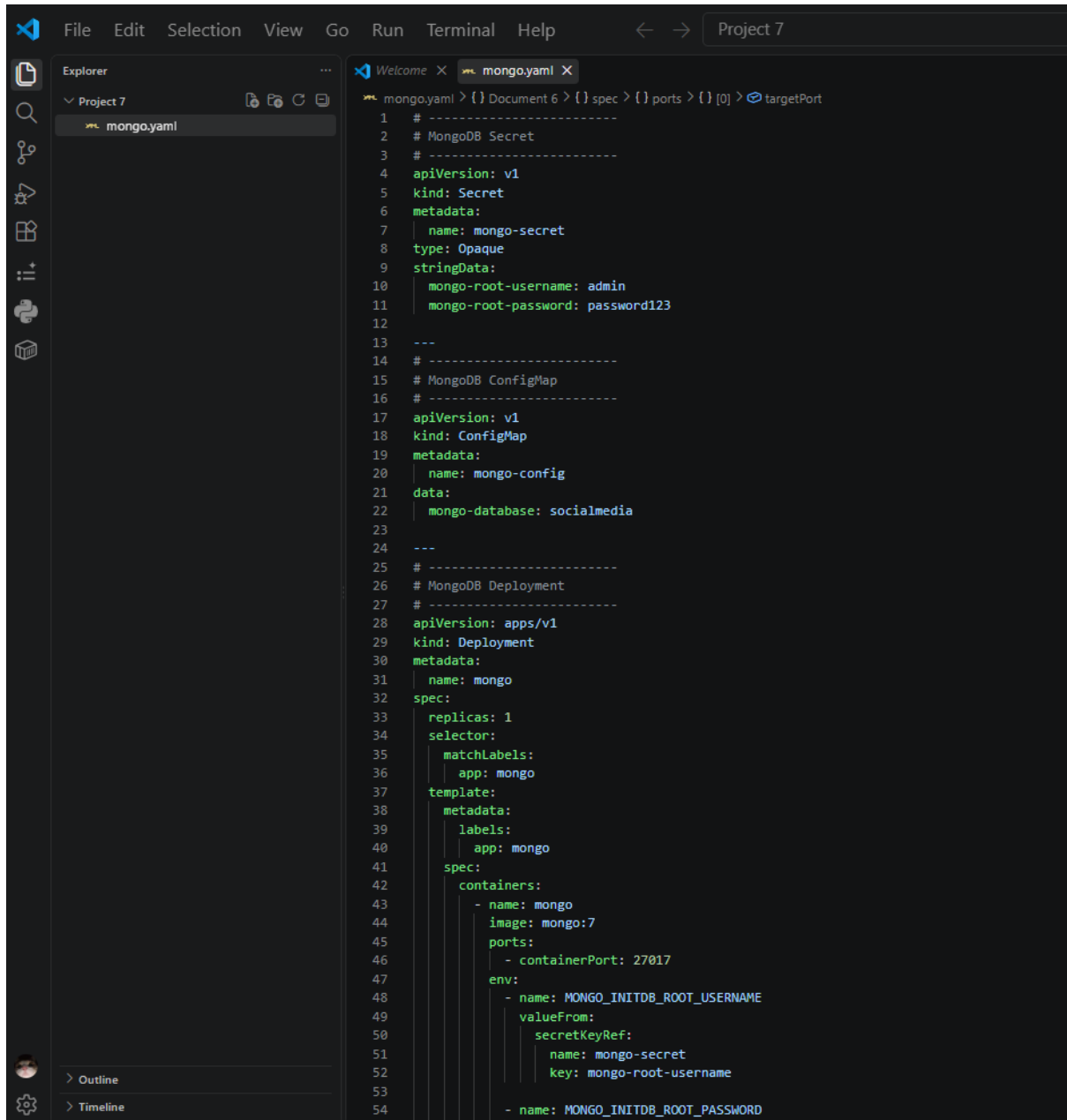


Name: Omkar Kadam

PRN: 23070122158



```
1 # -----
2 # MongoDB Secret
3 # -----
4 apiVersion: v1
5 kind: Secret
6 metadata:
7   name: mongo-secret
8 type: Opaque
9 stringData:
10   mongo-root-username: admin
11   mongo-root-password: password123
12
13 ---
14 # -----
15 # MongoDB ConfigMap
16 # -----
17 apiVersion: v1
18 kind: ConfigMap
19 metadata:
20   name: mongo-config
21 data:
22   mongo-database: socialmedia
23
24 ---
25 # -----
26 # MongoDB Deployment
27 # -----
28 apiVersion: apps/v1
29 kind: Deployment
30 metadata:
31   name: mongo
32 spec:
33   replicas: 1
34   selector:
35     matchLabels:
36       app: mongo
37   template:
38     metadata:
39       labels:
40         app: mongo
41     spec:
42       containers:
43         - name: mongo
44           image: mongo:7
45           ports:
46             - containerPort: 27017
47           env:
48             - name: MONGO_INITDB_ROOT_USERNAME
49               valueFrom:
50                 secretKeyRef:
51                   name: mongo-secret
52                   key: mongo-root-username
53             - name: MONGO_INITDB_ROOT_PASSWORD
```

File Edit Selection View Go Run Terminal Help ← → Project 7

Explorer

Project 7

mongo.yaml

mongo.yaml > {} Document 6 > {} spec > {} ports > {} [0] > targetPort

28 apiVersion: apps/v1
32 spec:
37 template:
41 spec:
42 containers:
55 valueFrom:
56 secretKeyRef:
57 name: mongo-secret
58 key: mongo-root-password
59
60 ---
61 # -----
62 # MongoDB Service
63 # -----
64 apiVersion: v1
65 kind: Service
66 metadata:
67 name: mongo-service
68 spec:
69 selector:
70 app: mongo
71 ports:
72 - port: 27017
73 targetPort: 27017
74
75 ---
76 # -----
77 # Mongo Express Deployment
78 # -----
79 apiVersion: apps/v1
80 kind: Deployment
81 metadata:
82 name: mongo-express
83 spec:
84 replicas: 1
85 selector:
86 matchLabels:
87 app: mongo-express
88 template:
89 metadata:
90 labels:
91 app: mongo-express
92 spec:
93 containers:
94 - name: mongo-express
95 image: mongo-express:1.0.2
96 ports:
97 - containerPort: 8081
98 env:
99 - name: ME_CONFIG_MONGODB_SERVER
100 value: mongo-service
101 - name: ME_CONFIG_MONGODB_PORT
102 value: "27017"
103

Outline
Timeline

File Edit Selection View Go Run Terminal Help ← → Project 7

Explorer

Project 7

mongo.yaml

Welcome X mongo.yaml X

mongo.yaml > {} Document 6 > {} spec > {} ports > {} [0] > targetPort

79 apiVersion: apps/v1

83 spec:

88 template:

92 spec:

93 containers:

102 - name: ME_CONFIG_MONGODB_PORT

103 value: "27017"

104

105 - name: ME_CONFIG_MONGODB_ADMINUSERNAME

106 valueFrom:

107 secretKeyRef:

108 name: mongo-secret

109 key: mongo-root-username

110

111 - name: ME_CONFIG_MONGODB_ADMINPASSWORD

112 valueFrom:

113 secretKeyRef:

114 name: mongo-secret

115 key: mongo-root-password

116

117 ---

118 # -----

119 # Mongo Express Service

120 # -----

121 apiVersion: v1

122 kind: Service

123 metadata:

124 name: mongo-express-service

125 spec:

126 type: NodePort

127 selector:

128 app: mongo-express

129 ports:

130 - port: 8081

131 targetPort: 8081

```

PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl apply -f mongo.yaml
secret/mongo-secret created
configmap/mongo-config created
deployment.apps/mongo created
service/mongo-service created
deployment.apps/mongo-express created
service/mongo-express-service created
PS Z:\0m\SIT\Semester 7\DevOps Lab\Project 7> kubectl get all

```

NAME	READY	STATUS	RESTARTS	AGE
pod/backend-54c679dfd9-g6hqb	1/1	Running	1 (33m ago)	24d
pod/flask-deployment-84964d4d6b-2p662	1/1	Running	2 (33m ago)	35d
pod/flask-deployment-84964d4d6b-lbncm	1/1	Running	2 (33m ago)	35d
pod/frontend-cbb6d74c4-cz2bx	1/1	Running	1 (33m ago)	24d
pod/load-generator	1/1	Running	0	4m1s
pod/mongo-58579ff459-mtp4f	0/1	ContainerCreating	0	7s
pod/mongo-express-5c6786d87f-t7t57	0/1	ContainerCreating	0	6s
pod/redis-696c967b87-d7qx2	1/1	Running	1 (33m ago)	24d
pod/social-media-app-5475bb6f57-txnkc	1/1	Running	0	2m27s
pod/social-media-app-5475bb6f57-znnd8	1/1	Running	0	28m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/backend-service	ClusterIP	10.99.250.180	<none>	5000/TCP	24d
service/flask-service	NodePort	10.109.217.59	<none>	5000:30007/TCP	35d
service/frontend-service	NodePort	10.103.10.173	<none>	80:31777/TCP	24d
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	35d
service/mongo-express-service	NodePort	10.108.217.44	<none>	8081:31340/TCP	7s
service/mongo-service	ClusterIP	10.110.253.176	<none>	27017/TCP	7s
service/redis	ClusterIP	10.99.67.140	<none>	6379/TCP	24d
service/social-media-service	NodePort	10.101.35.224	<none>	80:30405/TCP	28m

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/backend	1/1	1	1	24d
deployment.apps/flask-deployment	2/2	2	2	35d
deployment.apps/frontend	1/1	1	1	24d
deployment.apps/mongo	0/1	1	0	7s
deployment.apps/mongo-express	0/1	1	0	7s
deployment.apps/redis	1/1	1	1	24d
deployment.apps/social-media-app	2/2	2	2	28m

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/backend-54c679dfd9	1	1	1	24d
replicaset.apps/backend-66dd9756bc	0	0	0	24d
replicaset.apps/flask-deployment-84964d4d6b	2	2	2	35d
replicaset.apps/frontend-68756679cb	0	0	0	24d

PS Z:\Om\SIT\Semester 7\DevOps Lab\Project 7> minikube service mongo-express-service

NAMESPACE	NAME	TARGET PORT	URL
default	mongo-express-service	8081	http://192.168.49.2:31340

🔗 Starting tunnel for service mongo-express-service.

NAMESPACE	NAME	TARGET PORT	URL
default	mongo-express-service		http://127.0.0.1:61014

🌐 Opening service default/mongo-express-service in default browser...

! Because you are using a Docker driver on windows, the terminal needs to be open to run it.



